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Periodic Table & Periodicity

JEE Main Chemistry · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 182 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Open solution

Match List - I with List - II.

List - I Electronic configuration of neutral atom (where $n = 2$)		List - II 1st Ionization Energy ($kJ\ mol^{-1}$)	
A.	ns^2	I.	2080
B.	ns^2np^1	II.	899
C.	ns^2np^3	III.	800
D.	ns^2np^6	IV.	1402

Choose the correct answer from the options given below :

(A) A-II, B-III, C-IV, D-I

(B) A-IV, B-III, C-II, D-I

(C) A-III, B-II, C-IV, D-I

(D) A-III, B-II, C-I, D-IV

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

In a period, the first ionisation enthalpy of the element at extreme left and the negative electron gain enthalpy of the extreme right element, except noble gases, are respectively.

(A) lowest and lowest

(B) highest and lowest

(C) lowest and highest

(D) highest and highest

Q3. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Given below are two statements :

Statement I : The number of pairs among $[Al_2O_3, Cr_2O_3]$, $[Cl_2O_7, Mn_2O_7]$, $[Na_2O, V_2O_3]$ and $[CO, N_2O]$ that contain oxides of same nature (acidic, basic, neutral or amphoteric) is 4 .

Statement II : Among Na_2O, Al_2O_3, CO and Cl_2O_7, the most basic and acidic oxides are Na_2O and Cl_2O_7, respectively.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

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Q4. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Given below are two statements:

Statement I: The correct order of electronegativity of fluorine, oxygen and nitrogen is $F > O > N$.

Statement II: The oxidation state of oxygen in OF_2 is +2 and in Na_2O is -2 .

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q5. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

The 1st ionization enthalpy for Mg is $+737\ kJ/mol$. The most probable estimated value of the 2nd ionization enthalpy of Mg is ____ .

(A) $-906\ kJ/mol$

(B) $-856\ kJ/mol$

(C) $+1450\ kJ/mol$

(D) $+590\ kJ/mol$

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

The correct set that contains all kinds (basic, acidic, amphoteric and neutral) of oxides is :

(A) Na_2O, K_2O, Al_2O_3 and As_2O_3

(B) Al_2O_3, As_2O_3, CO and NO

(C) K_2O, Cl_2O_7, As_2O_3 and NO

(D) Na_2O, N_2O, Al_2O_3 and CO

Q7. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Given below are two statements :

Statement (I) : The first ionisation enthalpy of the elements Na, Mg, Cl and Ar follows the order Na > Mg > Cl > Ar.

Statement (II) : Among Ca, Al, Fe and B, the third ionisation enthalpy is very high for Ca.

In the light of the above statements, choose the **correct answer** from the options given below :

(A) Both **Statement I** and **Statement II** are true

(B) Both **Statement I** and **Statement II** are false

(C) **Statement I** is true but **Statement II** is false

(D) **Statement I** is false but **Statement II** is true

Q8. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Consider the elements N, P, O, S, Cl and F. The number of valence electrons present in the elements with most and least metallic character from the above list is respectively.

(A) 7 and 5

(B) 6 and 7

(C) 5 and 7

(D) 5 and 6

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Q9. JEE Main 2026 (Online) 28th January Morning Shift

In period 4 of the periodic table, the elements with highest and lowest atomic radii are respectively.

(A) Rb & Br (B) Na & Cl (C) $K \& Br$ (D) $K \& Se$

Open solution

Q10. JEE Main 2026 (Online) 24th January Evening Shift

The correct order of C, N, O and F in terms of second ionisation potential is

(A) $C < N < F < O$ (B) $C < F < N < O$

(C) $F < N < C < O$ (D) $C < O < N < F$

Open solution

Q11. JEE Main 2026 (Online) 24th January Morning Shift

Given below are two statements :

Statement I : $K > Mg > Al > B$ is the correct order in terms of metallic character.

Statement II : Atomic radius is always greater than the ionic radius for any element.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

Open solution

Q12. JEE Main 2026 (Online) 23rd January Evening Shift

Given below are two statements :

Statement I : The second ionisation enthalpy of Na is larger than the corresponding ionisation enthalpy of Mg .

Statement II : The ionic radius of O^{2-} is larger than that of F^{-}.

In the light of the above statements, choose the ***correct*** answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

Open solution

Q13. JEE Main 2026 (Online) 23rd January Evening Shift

Elements X and Y belong to Group 15. The difference between the electronegativity values of ' X ' and phosphorus is higher than that of the difference between phosphorus and ' Y '. ' X ' & ' Y ' are respectively

(A) As & Bi (B) Bi & N (C) N & As (D) $As \& Sb$

Open solution

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Q14. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

The correct trend in the first ionization enthalpies of the elements in the 3rd period of periodic table is :

(A) $Si < S < Al < P < Cl$

(B) $Al < Si < S < P < Cl$

(C) $S < Si < Al < P < Cl$

(D) $Al < S < P < Si < Cl$

Q15. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Given below are two statements :

Statement I : $C < O < N < F$ is the correct order in terms of first ionization enthalpy values.

Statement II : $S > Se > Te > Po > O$ is the correct order in terms of the magnitude of electron gain enthalpy values.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

Q16. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

Given below are two statements :

Statement I : The correct order in terms of atomic/ionic radii is Al > Mg > Mg^{2+} > Al^{3+}.

Statement II : The correct order in terms of the magnitude of electron gain enthalpy is Cl > Br > S > O.

In the light of the above statements, choose the correct answer from the options given below :

(A) **Statement I** is true but **Statement II** is false

(B) Both **Statement I** and **Statement II** are true

(C) **Statement I** is false but **Statement II** is true

(D) Both **Statement I** and **Statement II** are false

Q17. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Which of the following represents the correct trend for the mentioned property?

A. F > P > S > B - First Ionization Energy

B. Cl > F > S > P - Electron Affinity

C. K > Al > Mg > B - Metallic character

D. $K_2O > Na_2O > MgO > Al_2O_3$ - Basic character

Choose the **correct** answer from the options given below :

(A) B and C only (B) A, B and D only (C) A and B only (D) A, B, C and D

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Q18. JEE Main 2025 (Online) 8th April Evening Shift

The atomic number of the element from the following with lowest 1st ionisation enthalpy is :

(A) 32 (B) 35 (C) 19 (D) 87

Open solution

Q19. JEE Main 2025 (Online) 8th April Evening Shift

Given below are two statements:

Statement I: H_2Se is more acidic than H_2Te

Statement II: H_2Se has higher bond enthalpy for dissociation than H_2Te

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

Open solution

Q20. JEE Main 2025 (Online) 7th April Evening Shift

Choose the incorrect trend in the atomic radii (r) of the elements.

(A) $r_{Rb} < r_{Cs}$ (B) $r_{At} < r_{Cs}$ (C) $r_{Br} < r_{K}$ (D) $r_{Mg} < r_{Al}$

Open solution

Q21. JEE Main 2025 (Online) 7th April Morning Shift

The group 14 elements A and B have the first ionisation enthalpy values of 708 and 715 $\mathrm{kJ\ mol^{-1}}$ respectively. The above values are lowest among their group members. The nature of their ions $\mathrm{A^{2+}}$ and $\mathrm{B^{4+}}$ respectively is

(A) both reducing (B) oxidising and reducing

(C) both oxidising (D) reducing and oxidising

Open solution

Q22. JEE Main 2025 (Online) 4th April Evening Shift

The incorrect relationship in the following pairs in relation to ionisation enthalpies is :

(A) $\mathrm{Mn^{2+} < Fe^{2+}}$ (B) $\mathrm{Mn^{+} < Mn^{2+}}$

(C) $\mathrm{Mn^{+} < Cr^{+}}$ (D) $\mathrm{Fe^{2+} < Fe^{3+}}$

Open solution

Q23. JEE Main 2025 (Online) 4th April Evening Shift

The elements of Group 13 with highest and lowest first ionisation enthalpies are respectively :

(A) $\mathrm{B\ \&\ Tl}$ (B) $\mathrm{Tl\ \&\ B}$ (C) $B\ \&\ \mathrm{In}$ (D) $B\ \&\ Ga$

Open solution

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Q24. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

The correct orders among the following are

A. Atomic radius : $\mathrm{B} < \mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{Tl}$

B. Electronegativity : $\mathrm{Al} < \mathrm{Ga} < \mathrm{In} < \mathrm{Tl} < \mathrm{B}$

C. Density : $\mathrm{Tl} < \mathrm{In} < \mathrm{Ga} < \mathrm{Al} < \mathrm{B}$

D. 1st Ionisation Energy : $\mathrm{In} < \mathrm{Al} < \mathrm{Ga} < \mathrm{Tl} < \mathrm{B}$

Choose the correct answer from the options given below:

(A) B and D Only **(B)** A and B Only **(C)** C and D Only **(D)** A and C Only

Q25. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

Match the LIST-I with LIST-II

LIST-I (Family)		**LIST-II (Symbol of Element)**	
A.	Pnictogen (group 15)	I	Ts
B	Chalcogen	II	Og
C	Halogen	III	Lv
D	Noble gas	IV	Mc

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-II, D-III **(B)** A-IV, B-III, C-I, D-II

(C) A-III, B-I, C-IV, D-II **(D)** A-II, B-III, C-IV, D-I

Q26. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

Which of the following statements are correct?

A. The process of adding an electron to a neutral gaseous atom is always exothermic.

B. The process of removing an electron from an isolated gaseous atom is always endothermic.

C. The 1st ionization energy of boron is less than that of beryllium.

D. The electronegativity of C is 2.5 in $\mathrm{CH_4}$ and $\mathrm{CCl_4}$

E. Li is the most electropositive among elements of group I.

Choose the correct answer from the options given below:

(A) A, C and D Only **(B)** B and C Only **(C)** B and D Only **(D)** B, C and E Only

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Q27. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Electronic configuration of four elements A, B, C and D are given below :

(A) $1s^2 2s^2 2p^3$

(B) $1s^2 2s^2 2p^4$

(C) $1s^2 2s^2 2p^5$

(D) $1s^2 2s^2 2p^2$

Which of the following is the correct order of increasing electronegativity (Pauling's scale)?

(A) $\mathrm{D} < \mathrm{A} < \mathrm{B} < \mathrm{C}$ **(B)** $\mathrm{A} < \mathrm{B} < \mathrm{C} < \mathrm{D}$

(C) $\mathrm{A} < \mathrm{C} < \mathrm{B} < \mathrm{D}$ **(D)** $\mathrm{A} < \mathrm{D} < \mathrm{B} < \mathrm{C}$

Q28. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

Given below are two statements :

Statement (I) : The metallic radius of Al is less than that of Ga .

Statement (II) : The ionic radius of Al^{3+} is less than that of Ga^{3+}.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Both Statement I and Statement II are correct

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are incorrect

Q29. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

The property/properties that show irregularity in first four elements of group-17 is/are

(A) Covalent radius

(B) Electron affinity

(C) Ionic radius

(D) First ionization energy

Choose the correct answer from the options given below :

(A) A, B, C and D **(B)** A and C only **(C)** B only **(D)** B and D only

Q30. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

The type of oxide formed by the element among Li, Na, Be, Mg, B and Al that has the least atomic radius is :

(A) AO **(B)** AO_2 **(C)** A_2O **(D)** A_2O_3

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Q31. JEE Main 2025 (Online) 29th January Evening Shift

First ionisation enthalpy values of first four group 15 elements are given below. Choose the correct value for the element that is a main component of apatite family :

(A) 834 kJ mol^{-1} (B) 947 kJ mol^{-1} (C) 1402 kJ mol^{-1} (D) 1012 kJ mol^{-1}

Open solution

Q32. JEE Main 2025 (Online) 29th January Morning Shift

An element 'E' has the ionisation enthalpy value of 374 kJ mol^{-1}. 'E' reacts with elements A, B, C and D with electron gain enthalpy values of −328, −349, −325 and −295 kJ mol^{-1}, respectively. The correct order of the products EA, EB, EC and ED in terms of ionic character is :

(A) EB > EA > EC > ED
(B) EA > EB > EC > ED
(C) ED > EC > EA > EB
(D) ED > EC > EB > EA

Open solution

Q33. JEE Main 2025 (Online) 29th January Morning Shift

Given below are two statements :

Statement (I) : The radii of isoelectronic species increases in the order.

$Mg^{2+} < Na^{+} < F^{-} < O^{2-}$

Statement (II) : The magnitude of electron gain enthalpy of halogen decreases in the order.

Cl > F > Br > I

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is incorrect but Statement II is correct
(B) Both Statement I and Statement II are correct
(C) Statement I is correct but Statement II is incorrect
(D) Both Statement I and Statement II are incorrect

Open solution

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Q34. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Given below are two statements :

Statement I : According to the Law of Octaves, the elements were arranged in the increasing order of their atomic number.

Statement II : Meyer observed a periodically repeated pattern upon plotting physical properties of certain elements against their respective atomic numbers.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Both **Statement I** and **Statement II** are false

(B) **Statement I** is false but **Statement II** is true

(C) **Statement I** is true but **Statement II** is false

(D) Both **Statement I** and **Statement II** are true

Q35. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

The incorrect decreasing order of atomic radii is

(A) $\mathrm{Si} > \mathrm{P} > \mathrm{Cl} > \mathrm{F}$

(B) $\mathrm{Mg} > \mathrm{Al} > \mathrm{C} > \mathrm{O}$

(C) $\mathrm{Al} > \mathrm{B} > \mathrm{N} > \mathrm{F}$

(D) $\mathrm{Be} > \mathrm{Mg} > \mathrm{Al} > \mathrm{Si}$

Q36. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Consider the following elements $\mathrm{In}, \mathrm{Tl}, \mathrm{Al}, \mathrm{Pb}, \mathrm{Sn}$ and Ge . The most stable oxidation states of elements with highest and lowest first ionisation enthalpies, respectively, are

(A) +4 and +1 (B) +4 and +3 (C) +1 and +4 (D) +2 and +3

Q37. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

The successive 5 ionisation energies of an element are $800, 2427, 3658, 25024$ and $32824\ \mathrm{kJ/mol}$, respectively. By using the above values predict the group in which the above element is present :

(A) Group 4 (B) Group 2 (C) Group 13 (D) Group 14

Q38. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

Given below are two statements :

Statement (I) : The first ionization energy of Pb is greater than that of Sn .

Statement (II) : The first ionization energy of Ge is greater than that of Si .

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

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Q39. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

Which of the following statements are NOT true about the periodic table?

A. The properties of elements are function of atomic weights.

B. The properties of elements are function of atomic numbers.

C. Elements having similar outer electronic configurations are arranged in same period.

D. An element's location reflects the quantum numbers of the last filled orbital.

E. The number of elements in a period is same as the number of atomic orbitals available in energy level that is being filled.

Choose the correct answer from the options given below:

(A) A, C and E Only (B) A and E Only (C) B, C and E Only (D) D and E Only

Q40. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

The element that does not belong to the same period of the remaining elements (modern periodic table) is:

(A) Platinum (B) Osmium (C) Iridium (D) Palladium

Q41. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Given below are two statements :

Statement (I) : An element in the extreme left of the periodic table forms acidic oxides.

Statement (II) : Acid is formed during the reaction between water and oxide of a reactive element present in the extreme right of the periodic table.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

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Q42. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Match List-I with List-II.

	List - I		List - II
(A)	$Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$	(I)	Ionisation Enthalpy
(B)	$B < C < O < N$	(II)	Metallic character
(C)	$B < Al < Mg < K$	(III)	Electronegativity
(D)	$Si < P < S < Cl$	(IV)	Ionic radii

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(I), (C)- (II), (D)-(III)

(B) (A)-(III), (B)-(IV), (C)- (II), (D)-(I)

(C) (A)-(II), (B)-(III), (C)- (IV), (D)-(I)

(D) (A)-(IV), (B)-(I), (C)- (III), (D)-(II)

Q43. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Which of the following electronegativity order is incorrect?

(A) $S < Cl < O < F$

(B) $Al < Si < C < N$

(C) $Al < Mg < B < N$

(D) $Mg < Be < B < N$

Q44. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

Match List I with List II

	LIST I (Element)		LIST II (Electronic Configuration)
A.	N	I.	$[Ar]3\,d^{10}4\,s^{2}4p^{5}$
B.	S	II.	$[Ne]3\,s^{2}3p^{4}$
C.	Br	III.	$[He]2\,s^{2}2p^{3}$
D.	Kr	IV.	$[Ar]3\,d^{10}4\,s^{2}4p^{6}$

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-II, D-I

(B) A-I, B-IV, C-III, D-II

(C) A-III, B-II, C-I, D-IV

(D) A-II, B-I, C-IV, D-III

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Q45. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

Match List I with List II

	LIST I		LIST II
A.	Melting Point [K]	I.	$Tl > In > Ga > Al > B$
B.	Ionic Radius $[M^{+3}/pm]$	II.	$B > Tl > Al \approx Ga > In$
C.	$\Delta_i H_1 [kJ\ mol^{-1}]$	III.	$Tl > In > Al > Ga > B$
D.	Atomic Radius [pm]	IV.	$B > Al > Tl > In > Ga$

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-II, D-III

(B) A-I, B-II, C-III, D-IV

(C) A-III, B-IV, C-I, D-II

(D) A-II, B-III, C-IV, D-I

Q46. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

Given below are two statements :

Statement (I) : The oxidation state of an element in a particular compound is the charge acquired by its atom on the basis of electron gain enthalpy consideration from other atoms in the molecule.

Statement (II) : $p\pi - p\pi$ bond formation is more prevalent in second period elements over other periods.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Both Statement I and Statement II are correct

(C) Statement I is incorrect but Statement II is correct

(D) Both Statement I and Statement II are incorrect

Q47. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Match List I with List II

	LIST I (Elements)		LIST II (Properties in their respective groups)
A.	Cl, S	I.	Elements with highest electronegativity
B.	Ge, As	II.	Elements with largest atomic size
C.	Fr, Ra	III.	Elements which show properties of both metals and non-metal
D.	F, O	IV.	Elements with highest negative electron gain enthalpy

Choose the correct answer from the options given below:

(A) A-III, B-II, C-I, D-IV

(B) A-II, B-I, C-IV, D-III

(C) A-II, B-III, C-IV, D-I

(D) A-IV, B-III, C-II, D-I

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Q48. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

The electron affinity values are negative for

A. $Be \rightarrow Be^-$

B. $N \rightarrow N^-$

C. $O \rightarrow O^{2-}$

D. $Na \rightarrow Na^-$

E. $Al \rightarrow Al^-$

Choose the most appropriate answer from the options given below :

(A) A, B, D and E only

(B) D and E only

(C) A and D only

(D) A, B and C only

Q49. JEE Main 2024 (Online) 5th April Evening Shift

Open solution

Given below are two statements :

Statement I : The metallic radius of Na is $1.86\ A^\circ$ and the ionic radius of Na^+ is lesser than $1.86\ A^\circ$

Statement II : Ions are always smaller in size than the corresponding elements.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is incorrect but Statement II is true

(C) Both Statement I and Statement II are false

(D) Statement I is correct but Statement II is false

Q50. JEE Main 2024 (Online) 5th April Morning Shift

Open solution

The statement(s) that are correct about the species O^{2-}, F^-, Na^+ and Mg^{2+}.

(A) All are isoelectronic

(B) All have the same nuclear charge

(C) O^{2-} has the largest ionic radii

(D) Mg^{2+} has the smallest ionic radii

Choose the most appropriate answer from the options given below :

(A) (C) and (D) only

(B) (A), (C) and (D) only

(C) (B), (C) and (D) only

(D) (A), (B), (C) and (D)

Q51. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

The correct order of the first ionization enthalpy is

(A) $Ga > Al > B$

(B) $Tl > Ga > Al$

(C) $Al > Ga > Tl$

(D) $B > Al > Ga$

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Q52. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

Given below are two statements :

Statement I : The correct order of first ionization enthalpy values of $\mathrm{Li}, \mathrm{Na}, \mathrm{F}$ and Cl is $\mathrm{Na} < \mathrm{Li} < \mathrm{Cl} < \mathrm{F}$.

Statement II : The correct order of negative electron gain enthalpy values of $\mathrm{Li}, \mathrm{Na}, \mathrm{F}$ and Cl is $\mathrm{Na} < \mathrm{Li} < \mathrm{F} < \mathrm{Cl}$

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are true

Q53. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

The correct order of first ionization enthalpy values of the following elements is :

(A) O

(B) N

(C) Be

(D) F

(E) B

Choose the correct answer from the options given below :

(A) $\mathrm{A} < \mathrm{B} < \mathrm{D} < \mathrm{C} < \mathrm{E}$

(B) $\mathrm{C} < \mathrm{E} < \mathrm{A} < \mathrm{B} < \mathrm{D}$

(C) $\mathrm{E} < \mathrm{C} < \mathrm{A} < \mathrm{B} < \mathrm{D}$

(D) $\mathrm{B} < \mathrm{D} < \mathrm{C} < \mathrm{E} < \mathrm{A}$

Q54. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

Number of elements from the following that CANNOT form compounds with valencies which match with their respective group valencies is _____. B, C, N, S, O, F, P, Al, Si

(A) 7 **(B)** 3 **(C)** 5 **(D)** 6

Q55. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

The transition metal having highest 3rd ionisation enthalpy is :

(A) Mn **(B)** Fe **(C)** Cr **(D)** V

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Q56. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

Given below are two statements :

Statement (I) : Both metals and non-metals exist in p and d-block elements.

Statement (II) : Non-metals have higher ionisation enthalpy and higher electronegativity than the metals.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q57. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

In case of isoelectronic species the size of F^-, Ne and Na^+ is affected by :

(A) Nuclear charge (z)

(B) None of the factors because their size is the same

(C) Electron-electron interaction in the outer orbitals

(D) Principal quantum number (n)

Q58. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

Consider the following elements.

Group	A' B' → Period
↓	C' D'

Which of the following is/are true about A', B', C' and D' ?

A. Order of atomic radii: $B' < A' < D' < C'$

B. Order of metallic character: $B' < A' < D' < C'$

C. Size of the element: $D' < C' < B' < A'$

D. Order of ionic radii: $B'^{+} < A'^{+} < D'^{+} < C'^{+}$

Choose the correct answer from the options given below :

(A) A only (B) B, C and D only (C) A and B only (D) A, B and D only

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Q59. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

The correct sequence of electron gain enthalpy of the elements listed below is

A. Ar

B. Br

C. F

D. S

Choose the **most appropriate** from the options given below:

(A) $\mathrm{A} > \mathrm{D} > \mathrm{C} > \mathrm{B}$

(B) $\mathrm{A} > \mathrm{D} > \mathrm{B} > \mathrm{C}$

(C) $\mathrm{D} > \mathrm{C} > \mathrm{B} > \mathrm{A}$

(D) $\mathrm{C} > \mathrm{B} > \mathrm{D} > \mathrm{A}$

Q60. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: $\mathrm{H_2Te}$ is more acidic than $\mathrm{H_2\ S}$.

Reason R: Bond dissociation enthalpy of $\mathrm{H_2Te}$ is lower than $\mathrm{H_2\ S}$.

In the light of the above statements, choose the most appropriate from the options given below:

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is false but R is true.

(D) A is true but R is false.

Q61. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

Given below are two statements:

Statement - I: Along the period, the chemical reactivity of the elements gradually increases from group 1 to group 18 .

Statement - II: The nature of oxides formed by group 1 elements is basic while that of group 17 elements is acidic.

In the light of the above statements, choose the most appropriate from the options given below:

(A) Statement I is False but Statement II is true

(B) Both Statement I and Statement II are False

(C) Statement I is True But Statement II is False

(D) Both Statement I and Statement II are True

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Q62. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

Given below are two statements:

Statement I : Fluorine has most negative electron gain enthalpy in its group.

Statement II : Oxygen has least negative electron gain enthalpy in its group.

In the light of the above statements, choose the most appropriate from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q63. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

The element having the highest first ionization enthalpy is

(A) C (B) Al (C) Si (D) N

Q64. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

Anomalous behavior of oxygen is due to its

(A) large size and low electronegativity

(B) small size and high electronegativity

(C) small size and low electronegativity

(D) large size and high electronegativity

Q65. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$:

Assertion A : The first ionisation enthalpy decreases across a period.

Reason $\mathbf{R}$: The increasing nuclear charge outweighs the shielding across the period.

In the light of the above statements, choose the most appropriate from the options given below :

(A) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(D) $\mathbf{A}$ is true but $\mathbf{R}$ is false

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Q66. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Given below are two statements :

Statement I : The electronegativity of group 14 elements from Si to Pb, gradually decreases.

Statement II : Group 14 contains non-metallic, metallic, as well as metalloid elements.

In the light of the above statements, choose the **most appropriate** from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is true but Statement II is false

Q67. JEE Main 2024 (Online) 27th January Morning Shift

Open solution

Given below are two statements :

Statement (I) : The $4\mathrm{f}$ and $5\mathrm{f}$ - series of elements are placed separately in the Periodic table to preserve the principle of classification.

Statement (II) : S-block elements can be found in pure form in nature.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Q68. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

Which of the following statements are not correct?

A. The electron gain enthalpy of F is more negative than that of Cl.

B. Ionization enthalpy decreases in a group of periodic table.

C. The electronegativity of an atom depends upon the atoms bonded to it.

D. $\mathrm{Al_2O_3}$ and NO are examples of amphoteric oxides.

Choose the most appropriate answer from the options given below :

(A) A, B, C and D (B) A, C and D Only (C) A, B and D Only (D) B and D Only

Q69. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

For elements $\mathrm{B, C, N, Li, Be, O}$ and F, the correct order of first ionization enthalpy is

(A) $\mathrm{Li < Be < B < C < O < N < F}$

(B) $\mathrm{B > Li > Be > C > N > O > F}$

(C) $\mathrm{Li < Be < B < C < N < O < F}$

(D) $\mathrm{Li < B < Be < C < O < N < F}$

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Q70. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : The energy required to form Mg^{2+} from Mg is much higher than that required to produce Mg^{+}

Reason $\mathbf{R}$: Mg^{2+} is a small ion and carries more charge than Mg^{+}

In the light of the above statements, choose the correct answer from the options given below.

(A) A is false but R is true

(B) Both A and R are correct but R is NOT the correct explanation of A

(C) Both A and R are correct and R is the correct explanation of A

(D) A is true but R is false

Q71. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

The correct order of electronegativity for given elements is:

(A) $C > P > At > Br$

(B) $Br > C > At > P$

(C) $Br > P > At > C$

(D) $P > Br > C > At$

Q72. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

Which one of the following elements will remain as liquid inside pure boiling water?

(A) Li

(B) Br

(C) Cs

(D) Ga

Q73. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

Group-13 elements react with O_2 in amorphous form to form oxides of type M_2O_3 (M = element). Which among the following is the most basic oxide?

(A) Al_2O_3

(B) Tl_2O_3

(C) B_2O_3

(D) Ga_2O_3

Q74. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

The difference between electron gain enthalpies will be maximum between :

(A) Ar and Cl

(B) Ne and Cl

(C) Ne and F

(D) Ar and F

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Q75. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

For electron gain enthalpies of the elements denoted as $\Delta_{eg}H$, the incorrect option is :

(A) $\Delta_{eg}H(I) < \Delta_{eg}H(At)$

(B) $\Delta_{eg}H(Te) < \Delta_{eg}H(Po)$

(C) $\Delta_{eg}H(Cl) < \Delta_{eg}H(F)$

(D) $\Delta_{eg}H(Se) < \Delta_{eg}H(S)$

Q76. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

Assertion (A): The first ionization enthalpy of $3\,d$ series elements is more than that of group 2 metals

Reason (R): In 3d series of elements successive filling of d-orbitals takes place.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) (A) is true but (R) is false

(C) Both (A) and (R) are true but $(\mathbf{R})$ is not the correct explanation of (A)

(D) (A) is false but (R) is true

Q77. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

The correct increasing order of the ionic radii is

(A) $Cl^- < Ca^{2+} < K^+ < S^{2-}$

(B) $K^+ < S^{2-} < Ca^{2+} < Cl^-$

(C) $Ca^{2+} < K^+ < Cl^- < S^{2-}$

(D) $S^{2-} < Cl^- < Ca^{2+} < K^+$

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Q78. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

Bond dissociation energy of "E-H" bond of the "H_2E " hydrides of group 16 elements (given below), follows order.

A. O

B. S

C. Se

D. Te

Choose the correct answer from the options given below:

(A) $A > B > C > D$

(B) $D > C > B > A$

(C) $B > A > C > D$

(D) $A > B > D > C$

Q79. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

Match List I with List II

	List I (Atomic number)		List II (Block of periodic table)
A.	37	I.	p-block
B.	78	II.	d-block
C.	52	III.	f-block
D.	65	IV.	s-block

Choose the correct answer from the options given below :

(A) A - II, B - IV, C - I, D - III

(B) A - IV, B - III, C - II, D - I

(C) A - IV, B - II, C - I, D - III

(D) A - I, B - III, C - IV, D - II

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Q80. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

Given below are two statements :

Statement I : The decrease in first ionization enthalpy from B to Al is much larger than that from Al to Ga.

Statement II : The d orbitals in Ga are completely filled.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Statement I is correct but statement II is incorrect

(B) Statement I is incorrect but statement II is correct

(C) Both the statements I and II are correct

(D) Both the statements I and II are incorrect

Q81. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

Inert gases have positive electron gain enthalpy. Its correct order is :

(A) He < Ne < Kr < Xe

(B) He < Xe < Kr < Ne

(C) Xe < Kr < Ne < He

(D) He < Kr < Xe < Ne

Q82. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

For Periodic Table which of the following statements are TRUE

A. Shielding increases as we go down in the group.

B. Shielding increases sharply across the period.

C. Ionization enthalpy increases down the group.

D. Metallic character increases down the group.

E. Electronegativity decreases down the group.

Choose the correct answer from the following options :

(A) B, C and E Only

(B) A, D, and E only

(C) A, C, D and E only

(D) A, D and E only

Q83. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

The first ionization enthalpy of Na, Mg and Si, respectively, are : 496, 737 and $786\ \mathrm{kJ\ mol^{-1}}$. The first ionization enthalpy ($\mathrm{kJ\ mol^{-1}}$) of Al is :

(A) 487

(B) 768

(C) 577

(D) 856

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Q84. JEE Main 2022 (Online) 28th July Morning Shift

In which of the following pairs, electron gain enthalpies of constituent elements are nearly the same or identical?

(A) Rb and Cs

(B) Na and K

(C) Ar and Kr

(D) I and At

Choose the correct answer from the options given below :

(A) (A) and (B) only **(B)** (B) and (C) only **(C)** (A) and (C) only **(D)** (C) and (D) only

Open solution

Q85. JEE Main 2022 (Online) 27th July Evening Shift

Outermost electronic configurations of four elements A, B, C, D are given below :

(A) $3s^2$

(B) $3s^2 3p^1$

(C) $3s^2 3p^3$

(D) $3s^2 3p^4$

The correct order of first ionization enthalpy for them is :

(A) (A) < (B) < (C) < (D) **(B)** (B) < (A) < (D) < (C)

(C) (B) < (D) < (A) < (C) **(D)** (B) < (A) < (C) < (D)

Open solution

Q86. JEE Main 2022 (Online) 27th July Morning Shift

The incorrect statement is

(A) The first ionization enthalpy of K is less than that of Na and Li.

(B) Xe does not have the lowest first ionization enthalpy in its group.

(C) The first ionization enthalpy of element with atomic number 37 is lower than that of the element with atomic number 38.

(D) The first ionization enthalpy of Ga is higher than that of the d-block element with atomic number 30.

Open solution

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Q87. JEE Main 2022 (Online) 26th July Morning Shift

Open solution

Given two statements below :

Statement I : In Cl_2 molecule the covalent radius is double of the atomic radius of chlorine.

Statement II : Radius of anionic species is always greater than their parent atomic radius.

Choose the most appropriate answer from options given below :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q88. JEE Main 2022 (Online) 25th July Evening Shift

Open solution

The first ionization enthalpies of Be, B, N and O follow the order

(A) O < N < B < Be (B) Be < B < N < O (C) B < Be < N < O (D) B < Be < O < N

Q89. JEE Main 2022 (Online) 25th July Morning Shift

Open solution

The IUPAC nomenclature of an element with electronic configuration [Rn] $5f^{14}6d^{1}7s^{2}$ is :

(A) Unnilbium (B) Unnilunium (C) Unnilquadium (D) Unniltrium

Q90. JEE Main 2022 (Online) 30th June Morning Shift

Open solution

The correct order of electron gain enthalpy ($-$ ve value) is :

(A) O > S > Se > Te (B) O < S < Se < Te (C) O < S > Se > Te (D) O < S > Se < Te

Q91. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : The first ionization enthalpy for oxygen is lower than that of nitrogen.

Reason R : The four electrons in 2p orbitals of oxygen experience more electron-electron repulsion.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both A and R are correct and R is the correct explanation of A.

(B) Both A and R are correct but R is NOT the correct explanation of A.

(C) A is correct but R is not correct.

(D) A is not correct but R is correct.

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Q92. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

Match List - I with List - II :

	List-I (Oxide)		List-II (Nature)
(A)	Cl_2O_7	(I)	Amphoteric
(B)	Na_2O	(II)	Basic
(C)	Al_2O_3	(III)	Neutral
(D)	N_2O	(IV)	Acidic

Choose the correct answer from the options given below :

(A) (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

(B) (A) - (IV), (B) - (II), (C) - (I), (D) - (III)

(C) (A) - (II), (B) - (IV), (C) - (III), (D) - (I)

(D) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

Q93. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

Among the following, basic oxide is :

(A) SO_3 **(B)** SiO_2 **(C)** CaO **(D)** Al_2O_3

Q94. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

Element "E" belongs to the period 4 and group 16 of the periodic table. The valence shell electron configuration of the element, which is just above "E" in the group is

(A) $3s^2, 3p^4$ **(B)** $3d^{10}, 4s^2, 4p^4$ **(C)** $4d^{10}, 5s^2, 5p^4$ **(D)** $2s^2, 2p^4$

Q95. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

The correct order of increasing ionic radii is

(A) $Mg^{2+} < Na^+ < F^- < O^{2-} < N^{3-}$

(B) $N^{3-} < O^{2-} < F^- < Na^+ < Mg^{2+}$

(C) $F^- < Na^+ < O^{2-} < Mg^{2+} < N^{3-}$

(D) $Na^+ < F^- < Mg^{2+} < O^{2-} < N^{3-}$

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Q96. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The ionic radii of O^{2-} and Mg^{2+} are same.

Reason (R) : Both O^{2-} and Mg^{2+} are isoelectronic species.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Q97. JEE Main 2022 (Online) 25th June Evening Shift

Open solution

The correct order of electron gain enthalpies of Cl, F, Te and Po is

(A) F < Cl < Te < Po **(B)** Po < Te < F < Cl **(C)** Te < Po < Cl < F **(D)** Cl < F < Te < Po

Q98. JEE Main 2022 (Online) 24th June Morning Shift

Open solution

Given below are the oxides:

Na_2O, As_2O_3, N_2O, NO and Cl_2O_7

Number of amphoteric oxides is :

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

Q99. JEE Main 2021 (Online) 31st August Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Metallic character decreases and non-metallic character increases on moving from left to right in a period.

Reason (R) : It is due to increase in ionisation enthalpy and decrease in electron gain enthalpy, when one moves from left to right in a period.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) (A) is false but (R) is true.

(B) (A) is true but (R) is false

(C) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(D) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

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Q100. JEE Main 2021 (Online) 27th August Evening Shift

The correct order of ionic radii for the ions, P^{3-}, S^{2-}, Ca^{2+}, K^{+}, Cl^{-} is :

(A) $P^{3-} > S^{2-} > Cl^{-} > K^{+} > Ca^{2+}$

(B) $Cl^{-} > S^{2-} > P^{3-} > Ca^{2+} > K^{+}$

(C) $P^{3-} > S^{2-} > Cl^{-} > Ca^{2+} > K^{+}$

(D) $K^{+} > Ca^{2+} > P^{3-} > S^{2-} > Cl^{-}$

Open solution

Q101. JEE Main 2021 (Online) 27th July Evening Shift

The correct order of first ionisation enthalpy is :

(A) Mg < S < Al < P (B) Mg < Al < S < P (C) Al < Mg < S < P (D) Mg < Al < P < S

Open solution

Q102. JEE Main 2021 (Online) 27th July Morning Shift

Match List - I with List - II :

	List - I		List - II
(a)	NaOH	(i)	Acidic
(b)	$Be(OH)_2$	(ii)	Basic
(c)	$Ca(OH)_2$	(iii)	Amphoteric
(d)	$B(OH)_3$		
(e)	$Al(OH)_3$		

Open solution

Choose the most appropriate answer from the options given below

(A) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(ii), (e)-(iii)

(B) (a)-(ii), (b)-(iii), (c)-(ii), (d)-(i), (e)-(iii)

(C) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(i), (e)-(iii)

(D) (a)-(ii), (b)-(i), (c)-(ii), (d)-(iii), (e)-(iii)

Q103. JEE Main 2021 (Online) 25th July Evening Shift

The ionic radii of F^{-} and O^{2-} respectively are 1.33$\overset{o}{A}$ and 1.4$\overset{o}{A}$, while the covalent radius of N is 0.74$\overset{o}{A}$.

Open solution

The correct statement for the ionic radius of N^{3-} from the following is :

(A) It is smaller than F^{-} and N

(B) It is bigger than O^{2-} and F^{-}

(C) It is bigger than F^{-} and N, but smaller than of O^{2-}

(D) It is smaller than O^{2-} and F^{-}, but bigger than of N

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Q104. JEE Main 2021 (Online) 25th July Morning Shift

The ionic radii of K^+, Na^+, Al^{3+} and Mg^{2+} are in the order:

(A) $Na^+ < K^+ < Mg^{2+} < Al^{3+}$

(B) $Al^{3+} < Mg^{2+} < K^+ < Na^+$

(C) $Al^{3+} < Mg^{2+} < Na^+ < K^+$

(D) $K^+ < Al^{3+} < Mg^{2+} < Na^+$

Open solution

Q105. JEE Main 2021 (Online) 22th July Evening Shift

Which one of the following statements for D.I. Mendeleeff, is incorrect?

(A) He authored the textbook - Principles of Chemistry

(B) At the time, he proposed Periodic Table of elements structure of atom was known.

(C) Element with atomic number 101 is named after him.

(D) He invented accurate barometer.

Open solution

Q106. JEE Main 2021 (Online) 18th March Evening Shift

The first ionization energy of magnesium is smaller as compared to that of elements X and Y, but higher than that of Z. The elements X, Y and Z, respectively, are :

(A) neon, sodium and chlorine

(B) argon, chlorine and sodium

(C) chlorine, lithium and sodium

(D) argon, lithium and sodium

Open solution

Q107. JEE Main 2021 (Online) 18th March Morning Shift

The ionic radius of Na^+ ions is 1.02 $\overset{o}{A}$. The ionic radii (in $\overset{o}{A}$) of Mg^{2+} and Al^{3+}, respectively, are

(A) 1.05 and 0.99

(B) 0.72 and 0.54

(C) 0.85 and 0.99

(D) 0.68 and 0.72

Open solution

Q108. JEE Main 2021 (Online) 17th March Morning Shift

The absolute value of the electron gain enthalpy of halogens satisfies :

(A) Cl > Br > F > I

(B) Cl > F > Br > I

(C) I > Br > Cl > F

(D) F > Cl > Br > I

Open solution

Q109. JEE Main 2021 (Online) 16th March Evening Shift

The characteristics of elements X, Y and Z with atomic numbers, respectively, 33, 53 and 83 are :

(A) X and Y are metalloids and Z is a metal.

(B) X and Z are non-metals and Y is a metalloid.

(C) X is a metalloid, Y is a non-metal and Z is a metal.

(D) X, Y and Z are metals.

Open solution

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Q110. JEE Main 2021 (Online) 16th March Evening Shift

Identify the elements X and Y using the ionisation energy values given below :

	Ionization energy (kJ/mol)	
	1st	2nd
X	495	4563
Y	731	1450

Open solution

(A) X = F; Y = Mg (B) X = Mg; Y = F (C) X = Na; Y = Mg (D) X = Mg; Y = Na

Q111. JEE Main 2021 (Online) 26th February Evening Shift

Which pair of oxides is acidic in nature?

(A) CaO, SiO_2 (B) B_2O_3, CaO (C) B_2O_3, SiO_2 (D) N_2O, BaO

Open solution

Q112. JEE Main 2021 (Online) 26th February Evening Shift

The correct order of electron gain enthalpy is :

(A) Te > Se > S > O (B) S > Se > Te > O (C) S > O > Se > Te (D) O > S > Se > Te

Open solution

Q113. JEE Main 2021 (Online) 26th February Morning Shift

Match List - I with List - II.

Open solution

	List I Electronic configuration		List II $\Delta_i H$ in kJ mol^{-1}
(a)	$1s^2 2s^2$	(i)	801
(b)	$1s^2 2s^2 2p^4$	(ii)	899
(c)	$1s^2 2s^2 2p^3$	(iii)	1314
(d)	$1s^2 2s^2 2p^1$	(iv)	1402

Choose the most appropriate answer from the options given below :

(A) (a) → (i), (b) → (iii), (c) → (iv), (d) → (ii)

(B) (a) → (iv), (b) → (i), (c) → (ii), (d) → (iii)

(C) (a) → (ii), (b) → (iii), (c) → (iv), (d) → (i)

(D) (a) → (i), (b) → (iv), (c) → (iii), (d) → (ii)

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Q114. JEE Main 2021 (Online) 25th February Evening Shift

The correct order of bond dissociation enthalpy of halogens is :

(A) $F_2 > Cl_2 > Br_2 > I_2$

(B) $Cl_2 > Br_2 > F_2 > I_2$

(C) $I_2 > Br_2 > Cl_2 > F_2$

(D) $Cl_2 > F_2 > Br_2 > I_2$

Open solution

Q115. JEE Main 2021 (Online) 24th February Morning Shift

Consider the elements Mg, Al, S, P and Si, the correct increasing order of their first ionization enthalpy is :

(A) Al < Mg < Si < S < P

(B) Al < Mg < S < Si < P

(C) Mg < Al < Si < S < P

(D) Mg < Al < Si < P < S

Open solution

Q116. JEE Main 2020 (Online) 5th September Evening Slot

The correct order of the ionic radii of
O^{2-}, N^{3-}, F^- , Mg^{2+}, Na^+ and Al^{3+} is :

(A) $N^{3-} < O^{2-} < F^- < Na^+ < Mg^{2+} < Al^{3+}$

(B) $N^{3-} < F^- < O^{2-} < Mg^{2+} < Na^+ < Al^{3+}$

(C) $Al^{3+} < Na^+ < Mg^{2+} < O^{2-} < F^- < N^{3-}$

(D) $Al^{3+} < Mg^{2+} < Na^+ < F^- < O^{2-} < N^{3-}$

Open solution

Q117. JEE Main 2020 (Online) 4th September Evening Slot

The process that is NOT endothermic in nature is :

(A) $Ar(g) + e^- \rightarrow Ar^-(g)$

(B) $H(g) + e^- \rightarrow H^-(g)$

(C) $Na(g) \rightarrow Na^+(g) + e^-$

(D) $O^-(g) + e^- \rightarrow O^{2-}(g)$

Open solution

Q118. JEE Main 2020 (Online) 4th September Morning Slot

The ionic radii of O^{2-}, F^-, Na^+ and Mg^{2+} are in the order :

(A) $F^- > O^{2-} > Na^+ > Mg^{2+}$

(B) $Mg^{2+} > Na^+ > F^- > O^{2-}$

(C) $O^{2-} > F^- > Mg^{2+} > Na^+$

(D) $O^{2-} > F^- > Na^+ > Mg^{2+}$

Open solution

Q119. JEE Main 2020 (Online) 3rd September Evening Slot

Among the statements (I – IV), the correct ones are:
(I) Be has smaller atomic radius compared to Mg.
(II) Be has higher ionization enthalpy than Al.
(III) Charge/radius ratio of Be is greater than that of Al.
(IV) Both Be and Al form mainly covalent compounds.

(A) (I), (II) and (IV)

(B) (II), (III) and (IV)

(C) (I), (III) and (IV)

(D) (I), (II) and (III)

Open solution

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Q120. JEE Main 2020 (Online) 3rd September Evening Slot

The five successive ionization enthalpies of an element are 800, 2427, 3658, 25024 and 32824 kJ mol^{-1}. The number of valence electrons in the element is :

(A) 2 (B) 3 (C) 4 (D) 5

Open solution

Q121. JEE Main 2020 (Online) 3rd September Morning Slot

The atomic number of the element unnilennium is :

(A) 109 (B) 102 (C) 119 (D) 108

Open solution

Q122. JEE Main 2020 (Online) 2nd September Evening Slot

Three elements X, Y and Z are in the 3rd period of the periodic table. The oxides of X, Y and Z, respectively, are basic, amphoteric and acidic. The correct order of the atomic numbers of X, Y and Z is :

(A) X < Z < Y (B) Y < X < Z (C) Z < Y < X (D) X < Y < Z

Open solution

Q123. JEE Main 2020 (Online) 2nd September Morning Slot

In general the property (magnitudes only) that show an opposite trend in comparison to other properties across a period is

(A) Electron gain enthalpy (B) Electronegativity

(C) Ionization enthalpy (D) Atomic radius

Open solution

Q124. JEE Main 2020 (Online) 9th January Evening Slot

The first and second ionisation enthalpies of a metal are 496 and 4560 kJ mol^{-1}, respectively. How many moles of HCl and H_2SO_4, respectively, will be needed to react completely with 1 mole of the metal hydroxide ?

(A) 1 and 2 (B) 1 and 0.5 (C) 1 and 1 (D) 2 and 0.5

Open solution

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Q125. JEE Main 2020 (Online) 9th January Morning Slot

B has a smaller first ionization enthalpy than Be. Consider the following statements :

(I) It is easier to remove 2p electron than 2s electron

(II) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be.

(III) 2s electron has more penetration power than 2p electron.

(IV) atomic radius of B is more than Be (Atomic number B = 5, Be = 4)

The correct statements are :

(A) (I), (III) and (IV)
(B) (II), (III) and (IV)
(C) (I), (II) and (IV)
(D) (I), (II) and (III)

Open solution

Q126. JEE Main 2020 (Online) 9th January Morning Slot

The acidic, basic and amphoteric oxides, respectively, are :

(A) Cl_2O, CaO, P_4O_{10}
(B) N_2O_3, Li_2O, Al_2O_3
(C) MgO, Cl_2O, Al_2O_3
(D) Na_2O, SO_3, Al_2O_3

Open solution

Q127. JEE Main 2020 (Online) 8th January Evening Slot

The increasing order of the atomic radii of the following elements is :-
(a) C (b) O
(c) F (d) Cl
(e) Br

(A) (a) < (b) < (c) < (d) < (e)
(B) (c) < (b) < (a) < (d) < (e)
(C) (b) < (c) < (d) < (a) < (e)
(D) (d) < (c) < (b) < (a) < (e)

Open solution

Q128. JEE Main 2020 (Online) 8th January Morning Slot

The third ionization enthalpy is minimum for :

(A) Ni **(B)** Co **(C)** Mn **(D)** Fe

Open solution

Q129. JEE Main 2020 (Online) 8th January Morning Slot

The first ionization energy (in kJ/mol) of Na, Mg, Al and Si respectively, are :

(A) 496, 577, 786, 737
(B) 496, 737, 577, 786
(C) 786, 737, 577, 496
(D) 496, 577, 737, 786

Open solution

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Q130. JEE Main 2020 (Online) 7th January Evening Slot

Within each pair of elements F & Cl, S & Se, and Li & Na, respectively, the elements that release more energy upon an electron gain are :

(A) F, S and Li (B) Cl, Se and Na (C) Cl, S and Li (D) F, Se and Na

Open solution

Q131. JEE Main 2020 (Online) 7th January Morning Slot

The electron gain enthalpy (in kJ/mol) of fluorine, chlorine, bromine and iodine, respectively are:

(A) -296, -325, -333 and -349 (B) -349, -333, -325 and -296

(C) -333, -349, -325 and -296 (D) -333, -325, -349 and -296

Open solution

Q132. JEE Main 2019 (Online) 12th April Evening Slot

Among the following, the energy of 2s orbital is lowest in :

(A) Li (B) H (C) Na (D) K

Open solution

Q133. JEE Main 2019 (Online) 12th April Evening Slot

In comparison to boron, beryllium has :

(A) lesser nuclear charge and greater first ionisation enthalpy

(B) greater nuclear charge and greater first ionisation enthalpy

(C) greater nuclear charge and lesser first ionisation enthalpy

(D) lesser nuclear charge and lesser first ionisation enthalpy

Open solution

Q134. JEE Main 2019 (Online) 12th April Morning Slot

The group number, number of valence electrons, and valency of an element with atomic number 15, respectively, are:

(A) 16, 6 and 3 (B) 15, 6 and 12 (C) 16, 5 and 2 (D) 15, 5 and 3

Open solution

Q135. JEE Main 2019 (Online) 9th April Morning Slot

The element having greatest difference between its first and second ionization energies, is :

(A) Ba (B) Ca (C) Sc (D) K

Open solution

Q136. JEE Main 2019 (Online) 8th April Evening Slot

The IUPAC symbol for the element with atomic number 119 would be :

(A) Uun (B) Une (C) Uue (D) Unh

Open solution

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Q137. JEE Main 2019 (Online) 8th April Morning Slot

The size of the iso-electronic species Cl^-, Ar and Ca^{2+} is affected by :

(A) electron-electron interaction in the outer orbitals

(B) Principal quantum number of valence shell

(C) nuclear charge

(D) azimuthal quantum number of valence shell

Open solution

Q138. JEE Main 2019 (Online) 12th January Morning Slot

The element with Z = 120 (not yet discovered) will be an/a -

(A) Alkaline earth metal

(B) Alkali metal

(C) Transition metal

(D) Inner transition metal

Open solution

Q139. JEE Main 2019 (Online) 11th January Evening Slot

The correct option with respect to the Pauling electronegativity values of the element is :

(A) Ga < Ge

(B) Si < Al

(C) Te > Se

(D) P > S

Open solution

Q140. JEE Main 2019 (Online) 11th January Morning Slot

The correct order of the atomic radii of C, Cs, Al, and S is :

(A) S < C < Al < Cs

(B) S < C < Cs < Al

(C) C < S < Al < Cs

(D) C < S < Cs < Al

Open solution

Q141. JEE Main 2019 (Online) 10th January Morning Slot

The electronegativity of aluminium is similar to :

(A) Beryllium

(B) Carbon

(C) Boron

(D) Lithium

Open solution

Q142. JEE Main 2019 (Online) 9th January Evening Slot

When the first electron gain enthalpy $(\Delta_{eg}H)$ of oxygen is $-$ 141 kJ/mol, its second electron gain enthalpy is :

(A) a more negative value than the first

(B) almost the same as that of the first

(C) negative, but less negative than the first

(D) a positive value

Open solution

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Q143. JEE Main 2019 (Online) 9th January Morning Slot

In general, the properties that decrease and increase down a group in the periodic table, respectively, are :

(A) Atomic Radius and Electronegativity

(B) Electron Gain Enthalpy and Electronegativity.

(C) Electronegativity and Atomic Radius.

(D) Electronegativity and Electron Gain Enthalpy.

Open solution

Q144. JEE Main 2018 (Online) 15th April Evening Slot

The correct order of electron affinity is :

(A) F > Cl > O (B) F > O > Cl (C) Cl > F > O (D) O > F > Cl

Open solution

Q145. JEE Main 2018 (Online) 15th April Morning Slot

For Na^+, Mg^{2+}, F^- and O^{2-}; the correct order of increasing ionic radii is :

(A) $O^{2-} < F^- < Na^+ < Mg^{2+}$ (B) $Na^+ < Mg^{2+} < F^- < O^{2-}$

(C) $Mg^{2+} < Na^+ < F^- < O^{2-}$ (D) $Mg^{2+} < O^{2-} < Na^+ < F^-$

Open solution

Q146. JEE Main 2017 (Online) 9th April Morning Slot

The electronic configuration with the highest ionization enthalpy is :

(A) [Ne] $3s^2\ 3p^1$ (B) [Ne] $3s^2\ 3p^2$

(C) [Ne] $3s^2\ 3p^3$ (D) [Ar] $3d^{10}\ 4s^2\ 4p^3$

Open solution

Q147. JEE Main 2017 (Online) 9th April Morning Slot

Which one of the following is an oxide?

(A) KO_2 (B) BaO_2 (C) SiO_2 (D) CsO_2

Open solution

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Q148. JEE Main 2017 (Online) 8th April Morning Slot

Consider the following ionization enthalpies of two elements 'A' and 'B' .

Open solution

Element	Ionization enthalpy (kJ/mol)		
	1st	2nd	3rd
A	899	1757	14847
B	737	1450	7731

Which of the following statements is **correct** ?

(A) Both 'A' and 'B' belong to group-1where 'B' comes below 'A'.

(B) Both 'A' and 'B' belong to group-1 where 'A' comes below 'B'.

(C) Both 'A' and 'B' belong to group-2 where 'B' comes below 'A'.

(D) Both 'A' and 'B' belong to group-2 where 'A' comes below 'B'.

Q149. JEE Main 2016 (Online) 10th April Morning Slot

The following statements concern elements in the periodic table. Which of the following is true ?

Open solution

(A) All the elements in Group 17 are gases.

(B) The Group 13 elements are all metals.

(C) Elements of Group 16 have lower ionization enthalpy values compared to those of Group 15 in the corresponding periods.

(D) For Group 15 elements, the stability of + 5 oxidation state increases down the group.

Q150. JEE Main 2015 (Online) 10th April Morning Slot

In the long form of the periodic table, the valence shell electronic configuration of $5s^2\ 5p^4$ corresponds to the element present in :

Open solution

(A) Group 16 and period 6

(B) Group 17 and period 5

(C) Group 16 and period 5

(D) Group 17 and period 6

Q151. JEE Main 2015 (Offline)

The ionic radii (in Å) of N^{3-}, O^{2-} and F^- are respectively:

Open solution

(A) 1.36, 1.71 and 1.40

(B) 1.71, 1.40 and 1.36

(C) 1.71, 1.36 and 1.40

(D) 1.36, 1.40 and 1.71

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Q152. JEE Main 2014 (Online) 19th April Morning Slot

Which one of the following has largest ionic radius ?

(A) Li^+ (B) O_2^{2-} (C) B^{3+} (D) F^-

Open solution

Q153. JEE Main 2014 (Online) 12th April Morning Slot

Similarity in chemical properties of the atoms of elements in a group of the Periodic table is most closely related to :

(A) atomic numbers (B) atomic masses

(C) number of principal energy levels (D) number of valence electrons

Open solution

Q154. JEE Main 2014 (Online) 12th April Morning Slot

Which of the following arrangements represents the increasing order (smallest to largest) of ionic radii of the given species O^{2-}, S^{2-},N^{3-}, P^{3-} ?

(A) $O^{2-} < N^{3-} < S^{2-} < P^{3-}$ (B) $O^{2-} < P^{3-} < N^{3-} < S^{2-}$

(C) $N^{3-} < O^{2-} < P^{3-} < S^{2-}$ (D) $N^{3-} < S^{2-} < O^{2-} < P^{3-}$

Open solution

Q155. JEE Main 2014 (Online) 11th April Morning Slot

Which of the following series correctly represents relations between the elements from X to Y ?

X → Y

(A) $_3Li \rightarrow {}_{19}K$ Ionization enthalpy increases

(B) $_9F \rightarrow {}_{35}Br$ Electron gain enthalpy with negative sign increases

(C) $_6C \rightarrow {}_{32}Ge$ Atomic radii increases

(D) $_{18}Ar \rightarrow {}_{54}Xe$ Noble character increases

Open solution

Q156. JEE Main 2013 (Online) 23th April Morning Slot

Which is the correct order of second ionization potential of C, N, O and F in the following?

(A) O > N > F > C (B) O > F > N > C (C) F > O > N > C (D) C > N > O > F

Open solution

Q157. JEE Main 2013 (Online) 22th April Morning Slot

The order of increasing sizes of atomic radii among the elements O, S, Se and As is :

(A) As < S < O < Se (B) Se < S < As < O (C) O < S < Se < As (D) O < S < As < Se

Open solution

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Q158. JEE Main 2013 (Online) 9th April Morning Slot

Electron gain enthalpy with negative sign of fluorine is less than that of chlorine due to :

(A) High ionization enthalpy of fluorine
(B) Smaller size of chlorine atom
(C) Smaller size of fluorine atom
(D) Bigger size of 2p orbital of fluorine

Open solution

Q159. JEE Main 2013 (Offline)

Which of the following represents the correct order of increasing first ionization enthalpy for Ca, Ba, S, Se and Ar?

(A) S < Se < Ca < Ba < Ar
(B) Ba < Ca < Se < S < Ar
(C) Ca < Ba < S < Se < Ar
(D) Ca < S < Ba < Se < Ar

Open solution

Q160. JEE Main 2013 (Offline)

The first ionization potential of Na is 5.1 eV. The value of electron gain enthalpy of Na+ will be:

(A) − 5.1 eV
(B) −10.2 eV
(C) + 2.55 eV
(D) − 2.55 eV

Open solution

Q161. AIEEE 2012 (Online) 26th May

Which of the following presents the correct order of second ionization enthalpies of C, N, O and F ?

(A) $F > O > N > C$
(B) $O > N > F > C$
(C) $C > N > O > F$
(D) $O > F > N > C$

Open solution

Q162. AIEEE 2012 (Online) 19th May

Which pair of elements with the given atomic numbers is expected to have similar properties?

(A) 40,72
(B) 20,36
(C) 10,28
(D) 11,12

Open solution

Q163. AIEEE 2012 (Online) 12th May

Which among the following elements has the highest first ionization enthalpy?

(A) Nitrogen
(B) Boron
(C) Carbon
(D) Oxygen

Open solution

Q164. AIEEE 2012 (Online) 7th May

The electron affinity of chlorine is $3.7\ \text{eV}$. 1 gram of chlorine is completely converted to Cl^- ion in a gaseous state. $(1\ \text{eV} = 23.06\ \text{kcal mol}^{-1})$. Energy released in the process is

(A) $4.8\ \text{kcal}$
(B) $7.2\ \text{kcal}$
(C) $8.2\ \text{kcal}$
(D) $2.4\ \text{kcal}$

Open solution

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Q165. AIEEE 2012

The increasing order of the ionic radii of the given isoelectronic species is :

(A) $Cl^-, Ca^{2+}, K^+, S^{2-}$

(B) $S^{2-}, Cl^-, Ca^{2+}, K^+$

(C) $Ca^{2+}, K^+, Cl^-, S^{2-}$

(D) $K^+, S^{2-}, Ca^{2+}, Cl^-$

Open solution

Q166. AIEEE 2011

Which one of the following order represents the correct sequence of the increasing basic nature of the given oxides ?

(A) $MgO < K_2O < Al_2O_3 < Na_2O$

(B) $Al_2O_3 < MgO < Na_2O < K_2O$

(C) $Na_2O < K_2O < MgO < Al_2O_3$

(D) $K_2O < Na_2O < Al_2O_3 < MgO$

Open solution

Q167. AIEEE 2010

The correct sequence which shows decreasing order of the ionic radii of the elements is

(A) $Al^{3+} > Mg^{2+} > Na^+ > F^- > O^{2-}$

(B) $Na^+ > Mg^{2+} > Al^{3+} > O^{2-} > F^-$

(C) $Na^+ > F^- > Mg^{2+} > O^{2-} > Al^{3+}$

(D) $O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$

Open solution

Q168. AIEEE 2009

In which of the following arrangements, the sequence is not strictly according to the property written against it

(A) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power

(B) HF < HCl < HBr < HI : increasing acid strength

(C) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

(D) B < C < O < N : increasing first ionization enthalpy

Open solution

Q169. AIEEE 2009

The set representing the correct order of ionic radius is :

(A) $Na^+ > Li^+ > Mg^{2+} > Be^{2+}$

(B) $Li^+ > Na^+ > Mg^{2+} > Be^{2+}$

(C) $Mg^{2+} > Be^{2+} > Li^+ > Na^+$

(D) $Li^+ > Be^{2+} > Na^+ > Mg^{2+}$

Open solution

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Q170. AIEEE 2006

Following statements regarding the periodic trends of chemical reactivity of the alkali metals and the halogens are given. Which of these statements gives the correct picture?

(A) The reactivity decreases in the alkali metals but increases in the halogens with increase in atomic number down the group

(B) In both the alkali metals and the halogens the chemical reactivity decreases with increase in atomic number down the group

(C) Chemical reactivity increases with increase in atomic number down the group in both the alkali metals and halogens

(D) In alkali metals the reactivity increases but in the halogens it decreases with increase in atomic number down the group

Open solution

Q171. AIEEE 2006

The increasing order of the first ionization enthalpies of the elements B, P, S and F (lowest first) is :

(A) F < S < P < B (B) P < S < B < F (C) B < P < S < F (D) B < S < P < F

Open solution

Q172. AIEEE 2005

Which of the following oxides is amphoteric in character?

(A) CaO (B) CO_2 (C) SiO_2 (D) SnO_2

Open solution

Q173. AIEEE 2005

In which of the following arrangements the order is NOT according to the property indicated against it?

(A) Li < Na < K < Rb Increasing metallic radius

(B) I < Br < F < Cl Increasing electron gain enthalpy (with negative sign)

(C) B < C < N < O Increasing first ionization enthalpy

(D) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$ Increasing ionic size

Open solution

Q174. AIEEE 2004

Among Al_2O_3, SiO_2, P_2O_3 and SO_2 the correct order of acid strength is

(A) $SO_2 < P_2O_3 < SiO_2 < Al_2O_3$ (B) $Al_2O_3 < SiO_2 < P_2O_3 < SO_2$

(C) $Al_2O_3 < SiO_2 < SO_2 < P_2O_3$ (D) $SiO_2 < SO_2 < Al_2O_3 < P_2O_3$

Open solution

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Q175. AIEEE 2004

Which one of the following ions has the highest value of ionic radius?

(A) O^{2-} (B) B^{3+} (C) Li^{+} (D) F^{-}

Open solution

Q176. AIEEE 2004

The formation of the oxide ion $O^{2-}(g)$ requires first an exothermic and then an endothermic step as shown below

$O_{(g)} + e^{-} = O^{-}_{(g)}$ $\Delta H^{o} = -142$ kJmol^{-1}

$O^{-}_{(g)} + e^{-} = O^{2-}_{(g)}$ $\Delta H^{o} = 844$ kJmol^{-1}

This is because

(A) O^{-} ion will tend to resist the addition of another electron

(B) Oxygen has high electron affinity

(C) Oxygen is more electronegative

(D) O^{-} ion has comparatively larger size than oxygen atom

Open solution

Q177. AIEEE 2003

According to the Periodic Law of elements, the variation in properties of elements is related to their

(A) nuclear masses (B) atomic numbers

(C) nuclear neutron-proton number ratios (D) atomic masses

Open solution

Q178. AIEEE 2003

Which one of the following is an amphoteric oxide?

(A) Na_2O (B) SO_2 (C) B_2O_3 (D) ZnO

Open solution

Q179. AIEEE 2003

The reduction in atomic size with increase in atomic number is a characteristic of elements of

(A) d - block (B) f - block

(C) Radioactive series (D) High atomic masses.

Open solution

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Q180. AIEEE 2003

The atomic numbers of Vanadium (V), Chromium (cr), Manganese (Mn) and Iron (Fe), respectively, $23, 24, 25$ and 26. Which one of these may be expected to have the higher second ionization enthalpy?

(A) Cr (B) Mn (C) Fe (D) V

Open solution

Q181. AIEEE 2002

The correct order of ionic radius is

(A) $Ce > Sm > Tb > Lu$ (B) $Lu > Tb > Sm > Ce$

(C) $Tb > Lu > Sm > Ce$ (D) $Sm > Tb > Lu > Ce$

Open solution

Q182. AIEEE 2002

Ce^{+3}, La^{+3}, Pm^{+3} and Yb^{+3} have ionic radii in the increasing order as

(A) $La^{+3} < Ce^{+3} < Pm^{+3} < Yb^{+3}$ (B) $Yb^{+3} < Pm^{+3} < Ce^{+3} < La^{+3}$

(C) $La^{+3} = Ce^{+3} < Pm^{+3} < Yb^{+3}$ (D) $Yb^{+3} < Pm^{+3} < La^{+3} < Ce^{+3}$

Open solution

Numerical 5 questions

Q183. JEE Main 2026 (Online) 6th April Morning Shift

First and second ionization enthalpies of lithium are $520\ \text{kJ mol}^{-1}$ and $7297\ \text{kJ mol}^{-1}$ respectively. Energy required to convert 3.5 mg lithium (g) into $\text{Li}^{2+}(\text{g})$ $\left[\text{Li}(\text{g}) \rightarrow \text{Li}^{2+}(\text{g})\right]$ is ____ kJmol^{-1}. (nearest integer)

[Molar mass of $\text{Li} = 7\ \text{g mol}^{-1}$]

Open solution

Q184. JEE Main 2024 (Online) 30th January Morning Shift

If IUPAC name of an element is "Unununnium" then the element belongs to nth group of Periodic table. The value of n is _____.

Open solution

Q185. JEE Main 2023 (Online) 29th January Evening Shift

Total number of acidic oxides among

$\text{N}_2\text{O}_3, \text{NO}_2, \text{N}_2\text{O}, \text{Cl}_2\text{O}_7, \text{SO}_2, \text{CO}, \text{CaO}, \text{Na}_2\text{O}$ and NO is _____.

Open solution

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Q186. JEE Main 2021 (Online) 24th February Morning Shift

Number of amphoteric compounds among the following is _____.

(A) BeO

(B) BaO

(C) $Be(OH)_2$

(D) $Sr(OH)_2$

Open solution

Q187. JEE Main 2020 (Online) 6th September Evening Slot

The atomic number of Unnilunium is _____.

Open solution

Out of syllabus 1 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q188. JEE Main 2016 (Offline) · **MCQ (Single Correct Answer)**

Which of the following atoms has the highest first ionization energy?

(A) Na (B) K (C) Sc (D) Rb

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **C**	3. **A**	4. **A**	5. **C**
6. **C**	7. **D**	8. **C**	9. **C**	10. **A**
11. **C**	12. **C**	13. **C**	14. **B**	15. **D**
16. **C**	17. **B**	18. **D**	19. **D**	20. **D**
21. **D**	22. **A**	23. **C**	24. **A**	25. **B**
26. **B**	27. **A**	28. **C**	29. **C**	30. **D**
31. **D**	32. **A**	33. **B**	34. **A**	35. **D**
36. **B**	37. **C**	38. **D**	39. **A**	40. **D**
41. **D**	42. **A**	43. **C**	44. **C**	45. **A**
46. **C**	47. **D**	48. **D**	49. **D**	50. **B**
51. **B**	52. **D**	53. **C**	54. **B**	55. **A**
56. **C**	57. **A**	58. **D**	59. **B**	60. **A**
61. **A**	62. **C**	63. **D**	64. **B**	65. **A**
66. **A**	67. **C**	68. **B**	69. **D**	70. **C**
71. **B**	72. **D**	73. **B**	74. **B**	75. **D**
76. **D**	77. **C**	78. **A**	79. **C**	80. **C**
81. **B**	82. **B**	83. **C**	84. **C**	85. **B**
86. **D**	87. **D**	88. **D**	89. **D**	90. **C**
91. **B**	92. **B**	93. **C**	94. **A**	95. **A**
96. **D**	97. **B**	98. **B**	99. **B**	100. **A**
101. **C**	102. **B**	103. **B**	104. **C**	105. **B**
106. **B**	107. **B**	108. **B**	109. **C**	110. **C**
111. **C**	112. **B**	113. **C**	114. **B**	115. **A**
116. **D**	117. **B**	118. **D**	119. **A**	120. **B**
121. **A**	122. **D**	123. **D**	124. **B**	125. **D**
126. **B**	127. **B**	128. **D**	129. **B**	130. **C**
131. **C**	132. **D**	133. **A**	134. **D**	135. **D**
136. **C**	137. **C**	138. **A**	139. **A**	140. **C**
141. **A**	142. **D**	143. **C**	144. **C**	145. **C**
146. **C**	147. **C**	148. **C**	149. **C**	150. **C**

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151. **B**	152. **B**	153. **D**	154. **A**	155. **C**
156. **B**	157. **C**	158. **C**	159. **B**	160. **A**
161. **D**	162. **A**	163. **A**	164. **D**	165. **C**
166. **B**	167. **D**	168. **C**	169. **A**	170. **D**
171. **D**	172. **D**	173. **C**	174. **B**	175. **A**
176. **A**	177. **B**	178. **D**	179. **D**	180. **A**
181. **A**	182. **B**			

Numerical

183. **4**	184. **11**	185. **4**	186. **2**	187. **101**

Out of syllabus

188. **C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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